The Periodic Table Recast as S_N Representation Theory on S^{3N-1}*

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The non-relativistic N-electron Schrödinger equation, after separation of the hyperradius, reduces to angular dynamics on the hypersphere S^{3N-1}. The angular kinetic energy is the SO($3N$) Casimir operator with eigenvalue $\mu_{\text{free}} = \nu(\nu + 3N - 2)/2$. Fermionic antisymmetry restricts the spatial wavefunction to a specific S_N irreducible representation—the conjugate of the spin Young diagram—whose first-column length λ_1 fixes the minimum angular quantum number $\nu = N - \lambda_1$. For all atoms with ground-state spin $S < N/2$, the spatial irrep has $\lambda_1 = 2$, giving the universal result $\nu = N - 2$ and $\mu_{\text{free}} = 2(N-2)(N-1)$. This formula, combined with the structure of the S_N irrep, yields a classification of atoms into five types—A (trivial), B (noble gas), C (alkali), D (alkaline earth), E (open shell)—that recasts the periodic table's group structure in representation-theoretic language. The periodic law corresponds to the repetition of the S_N irrep sequence C → D → E → B within each period. This classification maps onto, rather than predicts, the known periodic table structure; its value lies in providing a unified group-theoretic framework for a pattern that is usually described empirically. In the per-pair limit, μ_{free} contributes exactly 4 units of angular momentum per antisymmetric electron pair as $N \to \infty$. We show that the topological structure (ν, μ_{free}) is smooth through $Z = 1/\alpha \approx 137$ and that the Dirac instability is a *metric* singularity on S^{3N-1}, not a topological one.

INTRODUCTION

The periodic table of elements, established empirically by Mendeleev in 1869 [1], received its first theoretical justification from quantum mechanics through the shell model and the aufbau principle. The $2n^2$ orbital counting rule, the grouping of elements by valence shell, and the progression of ionization energies across a period are all consequences of the hydrogen-like Schrödinger equation and its multi-electron extensions.

Despite this success, the periodic law itself—the recurrence of chemical properties with increasing atomic number—has not been derived from a single mathematical principle. Shell-filling rules are empirical (the Madelung rule has exceptions), Hund's rules require separate justification, and the distinction between noble gases, alkali metals, and transition metals is described rather than predicted.

In this paper, we show that the periodic table's group structure can be recast in the language of S_N representation theory acting on the spatial part of the N-electron wavefunction on the hypersphere S^{3N-1}. The key results are:

1. The ground-state angular quantum number is universally $\nu = N - 2$ for atoms with spin $S < N/2$ (Sec.).
2. The Pauli centrifugal cost $\mu_{\text{free}} = 2(N-2)(N-1)$ grows quadratically, contributing exactly 4 units per electron pair in the large-N limit (Sec.).
3. The shape of the S_N irrep classifies atoms into chemical types that recast the group structure of the periodic table in representation-theoretic language (Sec.).
4. The periodic law corresponds to the repetition of S_N irrep sequences (Sec.).
5. The Dirac instability at $Z\alpha \to 1$ is a metric singularity, not a topological one; the representation theory is smooth through $Z = 137$ (Sec.).

WHAT IS NEW VS. WHAT IS KNOWN

Several ingredients of the present analysis are standard; it is important to distinguish them from the new contributions.

Known results

- **Hyperspherical formulation.** The separation of the N-electron Schrödinger equation into hyperradial and angular parts on S^{3N-1}, and the identification of the angular kinetic energy with the SO($3N$) Casimir, are established [3, 4].
- **S_N representation theory for fermions.** The use of Young diagrams and Schur–Weyl duality to classify spatial–spin factorizations of antisymmetric wavefunctions is textbook material [5, 6].
- **The Casimir eigenvalue formula.** The expression $\mu_{\text{free}} = \nu(\nu + 3N - 2)/2$ follows directly from the standard SO($3N$) representation theory.

- **The periodic table itself.** The classification of elements into noble gases, alkali metals, alkaline earths, and transition metals is an empirical fact that guided the construction presented here.

New contributions

1. **The universal $\nu = N - 2$ result.** Theorem 1 shows that, for all ground-state atoms with $S < N/2$, the spatial irrep has first-column length $\lambda_1 = 2$, yielding the universal angular quantum number $\nu = N - 2$ and $\mu_{\text{free}} = 2(N-2)(N-1)$. This specific identification appears to be new.
2. **The five-type classification (A/B/C/D/E).** The observation that the S_N irrep shape naturally partitions atoms into types that map onto the periodic table's group structure provides a group-theoretic reformulation of the empirical pattern.
3. **The per-pair angular momentum limit.** The result that $\mu_{\text{free}}/\binom{N}{2} \to 4$ as $N \to \infty$—each antisymmetric electron pair contributes exactly 4 units of angular momentum—connects the periodic pattern to a quantitative angular momentum budget.
4. **The Dirac instability as metric vs. topological singularity.** The analysis showing that ν, μ_{free}, and the S_N irrep are smooth through $Z = 137$ while the metric diverges clarifies the nature of the upper limit on the periodic table.

THE HYPERSPHERICAL FRAMEWORK

Configuration space and angular dynamics

Consider N electrons in a central potential $V(r_1, \ldots, r_N)$. After extracting the center-of-mass motion, the configuration space is $\mathbb{R}^{3N}$. Introducing hyperspherical coordinates $(R, \hat{\Omega})$ where $R = \sqrt{\sum_i r_i^2}$ is the hyperradius and $\hat{\Omega} \in S^{3N-1}$ collects $3N-1$ angular variables, the kinetic energy separates as [3, 4]:

$$T = -\frac{1}{2}\left(\frac{\partial^2}{\partial R^2} + \frac{3N-1}{R}\frac{\partial}{\partial R} - \frac{\Lambda^2}{R^2}\right), \tag{1}$$

where Λ^2 is the grand angular momentum operator—the Laplace–Beltrami operator on S^{3N-1}. Its eigenvalues are the SO($3N$) Casimir invariants:

$$\Lambda^2 Y_\nu = \nu(\nu + 3N - 2)\, Y_\nu \equiv 2\mu_{\text{free}}\, Y_\nu, \tag{2}$$

with $\nu = 0, 1, 2, \ldots$ and hyperspherical harmonics Y_ν. The effective angular momentum quantum numbers are

$$l_{\text{eff}} = \frac{3N-3}{2}, \quad n_{\text{eff}} = \frac{3N-1}{2}. \tag{3}$$

The centrifugal barrier in the hyperradial equation is $l_{\text{eff}}(l_{\text{eff}}+1)/R^2 = (3N-3)(3N-1)/(4R^2)$, showing that even in the lowest angular channel, N-electron atoms face a centrifugal cost that grows as $\sim 9N^2/4$.

Fermionic antisymmetry and S_N irreps

The total wavefunction must be antisymmetric under S_N (permutation of electrons). Since the total state factorizes as $|\Psi\rangle = |\text{spatial}\rangle \otimes |\text{spin}\rangle$, the spatial and spin parts must carry conjugate S_N irreducible representations.

For N spin-1/2 fermions with total spin S, the spin irrep is a Young diagram with at most two rows: $[n_\uparrow, n_\downarrow]$ where $n_\uparrow = N/2 + S$ and $n_\downarrow = N/2 - S$. By Schur–Weyl duality [5, 6], the spatial irrep is the *conjugate* (transposed) Young diagram.

Theorem 1 (Universal λ_1). *For any N-electron atom with ground-state spin $S < N/2$, the spatial Young diagram has first-column length $\lambda_1 = 2$.*

Proof. The spin diagram has two rows: $[n_\uparrow, n_\downarrow]$ with $n_\downarrow \geq 1$ (since $S < N/2$ implies $n_\downarrow > 0$). The conjugate diagram has $n_\downarrow$ columns of height 2 and $n_\uparrow - n_\downarrow = 2S$ columns of height 1. Therefore the first column has height $\lambda_1 = 2$, independent of S (provided $S < N/2$). □

The ground-state angular quantum number

The minimum ν compatible with a given S_N irrep is determined by the constraint that the hyperspherical harmonic Y_ν must transform in that irrep under particle permutations. For the spatial irrep with first-column length λ_1, the minimum angular excitation is [4]:

$$\nu_{\min} = N - \lambda_1. \tag{4}$$

By Theorem , all atoms with $S < N/2$ have

$$\boxed{\nu = N - 2} \tag{5}$$

as their ground-state angular quantum number. This is a *universal* result: it depends only on the electron count N, not on the nuclear charge Z, the spin S (provided $S < N/2$), or any details of the electron configuration.

The sole exception is the maximally polarized state $S = N/2$ (all spins parallel), which has $\lambda_1 = N$ and $\nu = 0$. This state is completely symmetric in space and carries zero angular momentum—it is the Laughlin-like state with no Pauli nodes.

THE μ_{free} FORMULA

Substituting $\nu = N-2$ into Eq. (2):

$$\mu_{\text{free}} = \frac{\nu(\nu+3N-2)}{2} = \frac{(N-2)(N-2+3N-2)}{2} = \frac{(N-2)(4N-4)}{2} = \boxed{2(N-2)(N-1)}. \quad (6)$$

This is the *Pauli centrifugal cost*: the minimum angular kinetic energy imposed by fermionic antisymmetry on S^{3N-1}.

Asymptotic behavior

Several limits of Eq. (6) are physically informative:

1. **Per-electron scaling.** $\mu_{\text{free}}/N^2 = 2 - 6/N + 4/N^2 \to 2$ as $N \to \infty$.

2. **Per-pair scaling.** The number of electron pairs is $\binom{N}{2} = N(N-1)/2$. The Pauli cost per pair is

$$\frac{\mu_{\text{free}}}{\binom{N}{2}} = \frac{4(N-2)}{N} \longrightarrow 4 \quad (N \to \infty). \quad (7)$$

   Each antisymmetric electron pair contributes exactly 4 units of angular momentum to the ground-state Casimir.

3. **Excitation fraction.** The ratio $\nu/(3N-2) = (N-2)/(3N-2) \to 1/3$ monotonically from below. The ground state always sits at approximately one-third of the natural angular scale of the sphere, regardless of N.

4. **Fractional ionization cost.** Removing one electron changes the sphere dimension by $3/(3N-1) \to 0$. For heavy atoms, ionization is a *small perturbation* to the angular topology.

Table I collects these quantities for atoms across the periodic table.

STRUCTURE TYPE CLASSIFICATION

While $\nu = N-2$ and $\mu_{\text{free}} = 2(N-2)(N-1)$ are universal, the *shape* of the S_N irrep varies with the electron configuration. This shape encodes the internal structure of the atom—how the N electrons partition into core and valence.

We define five structure types based on the spatial Young diagram:

TABLE I. Ground-state hyperspherical parameters for neutral atoms ($S < N/2$). $\nu = N-2$, $\mu_{\text{free}} = 2(N-2)(N-1)$, and the asymptotic ratios $\mu_{\text{free}}/N^2 \to 2$, $\mu_{\text{free}}/\binom{N}{2} \to 4$, $\nu/(3N-2) \to 1/3$.

Atom	N	S^{3N-1}	ν	μ_{free}	μ_{free}/N^2	Type
H	1	S^2	0	0	0.00	A
He	2	S^5	0	0	0.00	B
Li	3	S^8	1	4	0.44	C
Be	4	S^{11}	2	12	0.75	D
N	7	S^{20}	5	60	1.22	E
Ne	10	S^{29}	8	144	1.44	B
Na	11	S^{32}	9	180	1.49	C
Mg	12	S^{35}	10	220	1.53	D
Ar	18	S^{53}	16	544	1.68	B
Kr	36	S^{107}	34	2380	1.84	B
Xe	54	S^{161}	52	5512	1.89	B
Rn	86	S^{257}	84	14280	1.93	B
Og	118	S^{353}	116	27144	1.95	B

Type A: Trivial ($N = 1$)

The hydrogen atom has a single electron. The trivial irrep [1] of S_1 gives $\lambda_1 = 1$, $\nu = 0$, $\mu_{\text{free}} = 0$. There is no Pauli constraint. The wavefunction lives on S^2 (a single electron's angular space).

Type B: Noble gas (democratic structure)

Noble gases have closed-shell ground states with $S = 0$. The spin diagram is $[N/2, N/2]$ and the spatial irrep is its conjugate: $[2, 2, \ldots, 2]$ ($N/2$ rows of length 2).

This is the most *democratic* irrep: all columns have the same height. No electron is distinguished. The wavefunction has the full point-group symmetry of the Young diagram, and ionization requires breaking this symmetry—a costly operation reflected in the high ionization energies of noble gases (He: 24.6 eV, Ne: 21.6 eV, Ar: 15.8 eV).

Type C: Alkali metal (hierarchical structure)

Alkali metals have one valence electron outside a noble-gas core, with $S = 1/2$. The spin diagram is $[(N+1)/2, (N-1)/2]$ and the spatial irrep is $[2, 2, \ldots, 2, 1]$ ($= [2^{(N-1)/2}, 1]$).

This is a *hierarchical* irrep: one column (the valence electron) is shorter than the rest. The wavefunction has a natural decomposition into a symmetric core and a weakly bound valence electron. Ionization removes the short column—a topologically clean surgery on S^{3N-1}:

$$S^{3N-1} \to S^{3(N-1)-1} \times \mathbb{R}. \quad (8)$$

The low cost of this surgery explains the low ionization energies of alkali metals (Li: 5.4 eV, Na: 5.1 eV, K: 4.3 eV).

Type D: Alkaline earth (core + pair)

Alkaline earth metals have two valence electrons outside a noble-gas core, with $S = 0$. The spatial irrep is $[2, 2, \ldots, 2]$ (same shape as Type B) but the *occupancy pattern* distinguishes the core from the two valence electrons.

The ionization energy is intermediate (Be: 9.3 eV, Mg: 7.6 eV, Ca: 6.1 eV)—higher than alkali metals (the valence pair is harder to break) but lower than noble gases (the valence pair is still distinguishable from the core).

Type E: Open shell (partially filled p or d)

Atoms with a partially filled p shell (boron through fluorine) or d shell (the transition metals) carry an open valence shell with $S > 0$ but $S < N/2$. The spatial irrep is neither the democratic $[2^{N/2}]$ of Type B/D nor the single-defect $[2^{(N-1)/2}, 1]$ of Type C, but an intermediate shape with several columns of unequal height set by the open-shell occupancy and Hund's rule. This is the broadest class and the one where intra-shell exchange coupling is strongest; the composed-builder implementation refines it further into a p-block dispatch (type E) and a separate d-block dispatch (type F) because the angular machinery differs, but at the representation-theory level both are open-shell instances of the same intermediate-irrep family. Ionization energies span the widest range (B: 8.3 eV to F: 17.4 eV), tracking the filling of the open shell.

Table II summarizes the classification.

TABLE II. Structure type classification. The spatial Young diagram is the conjugate of the spin diagram. IE_1 is the first ionization energy (experimental).

Type	Spatial irrep	Chemistry	Examples	IE_1 range
A	$[1]$	Monatomic	H	13.6 eV
B	$[2^{N/2}]$	Noble gas	He, Ne, Ar	15–25 eV
C	$[2^{(N-1)/2}, 1]$	Alkali	Li, Na, K	4–5 eV
D	$[2^{N/2}]$	Alkaline earth	Be, Mg, Ca	6–9 eV
E	Various	Open shell	B–F, d-block	8–17 eV

THE PERIODIC LAW AS IRREP SEQUENCE

The periodic table's group structure arises because the sequence of S_N structure types *repeats* with each period. Within a given period (defined by the outermost principal quantum number n), the element sequence is:

$$\mathrm{C} \to \mathrm{D} \to \underbrace{\mathrm{E} \to \cdots \to \mathrm{E}}_{p\text{- and } d\text{-block}} \to \mathrm{B}. \tag{9}$$

Each period begins with an alkali metal (Type C: one electron in the new shell) and ends with a noble gas (Type B: closed shell). The intermediate elements (Type E) have partially filled subshells with varying spin and Young diagram shapes.

In group-theoretic language, the periodic law can be understood as follows:

> *The chemical properties of elements are periodic functions of atomic number because the sequence of S_N spatial irreps repeats whenever a new principal shell begins filling.*

This is a reformulation, not a derivation from first principles: the shell-filling order itself (which determines where the sequence repeats) is an empirical input, not a consequence of S_N representation theory alone.

The key distinction between Types B and D is worth emphasizing. Both have the same Young diagram shape $[2^{N/2}]$, but they differ in the *occupancy pattern*: Type B fills all available subshells, while Type D has two electrons in a new shell above a closed core. This distinction is invisible at Level 1 (representation theory) and becomes visible only at Level 2 (the perturbative energy, which depends on Z, orbital angular momenta, and electron–electron repulsion).

Isoelectronic series

Atoms with the same N but different Z share identical S_N irreps and structure types. The 10-electron series

$$\mathrm{Ne},\ \mathrm{Na}^{+},\ \mathrm{Mg}^{2+},\ \mathrm{Al}^{3+},\ \ldots \tag{10}$$

all have the Type B spatial irrep $[2, 2, 2, 2, 2]$ with $\nu = 8$ and $\mu_{\mathrm{free}} = 144$. Their chemistry differs only through the nuclear charge Z, which enters at Level 2. This cleanly separates *topology* (same for the series) from *potential* (varies with Z).

Topology flow: ionization as surgery

Ionization changes the electron count $N \to N - 1$, transforming the configuration space via Eq. (8). This is a codimension-3 surgery that removes three dimensions (one electron's $\mathbb{R}^3$) from the hypersphere. The fractional dimension loss is $3/(3N - 1)$, which decreases monotonically:

$$\frac{3}{3N-1} \to 0 \quad (N \to \infty). \tag{11}$$

For heavy atoms, ionization is a negligible topological perturbation. This correlates with the empirical observation that ionization energies decrease down a group: removing one electron from S^{353} (Og) is geometrically less disruptive than removing one from S^5 (He).

TOPOLOGY VERSUS METRIC: TWO-LEVEL STRUCTURE

The preceding analysis reveals a clean two-level separation:

- **Level 1 (Topological).** The quantities ν, μ_{free}, the S_N irrep, and the structure type depend only on the electron count N and the total spin S. They are independent of Z, the orbital angular momenta, and all details of the electron–electron interaction. They are *universal*.
- **Level 2 (Perturbative/Metric).** The binding energy, ionization energy, electron configuration, and Hund's rules all depend on Z, the fine-structure constant α, and the shape of the potential. These quantities determine the *chemistry* within a given topological class.

An important consequence: *Hund's first rule is not a topological effect.* Since $\lambda_1 = 2$ for all spatial irreps with $S < N/2$, the angular quantum number $\nu = N - 2$ is independent of the spin value. The preference for high-spin ground states (Hund's first rule) enters through the perturbative Coulomb and exchange integrals at Level 2, not through a reduction of ν at Level 1.

The Dirac limit as metric singularity

The Dirac equation for hydrogen-like atoms predicts that the 1s binding energy diverges at $Z\alpha = 1$ ($Z \approx 137$ for a point nucleus):

$$E_{1s}^{\text{Dirac}} = mc^2 \left(\sqrt{1 - (Z\alpha)^2} - 1 \right). \tag{12}$$

Does this "Dirac dive" have a topological signature on S^{3N-1}?

At $N = 137$: $\nu = 135$, $\mu_{\text{free}} = 36{,}720$, and $\nu/(3N - 2) = 0.3301$. All formulas are perfectly smooth. The excitation fraction approaches 1/3 from below with no singularity, no phase transition, and no special structure at $N = 137$.

The Dirac instability is instead a *metric* effect. The non-relativistic Fock projection yields the round metric on S^3. Relativistic corrections warp this metric by a conformal factor

$$\delta_{1s} = \frac{(Z\alpha)^2}{1 - (Z\alpha)^2}, \tag{13}$$

which diverges at $Z\alpha \to 1$. This "pinch" is a conical singularity at the nuclear pole of S^{3N-1}—the geometric avatar of the Dirac dive. The topology (S_N irrep, ν, μ_{free}) is oblivious; the metric (curvature, conformal factor) is singular.

Table III quantifies the transition from perturbative to singular.

TABLE III. Relativistic metric correction $\delta_{1s} = (Z\alpha)^2/[1 - (Z\alpha)^2]$ for the innermost electron. The non-relativistic μ_{free} formula becomes unreliable when $\delta \gtrsim 1$.

Z	$(Z\alpha)^2$	δ_{1s}	Regime
1	0.00005	0.00005	Non-relativistic
29	0.045	0.047	Mildly relativistic
47	0.118	0.133	Relativistic
79	0.332	0.498	Strongly relativistic
118	0.741	2.87	Near-singular
137	0.999	$\sim 10^3$	Conical singularity

Three limits on the periodic table

The periodic table terminates due to physics beyond the non-relativistic S_N representation theory:

1. **Nuclear instability** ($Z \sim 126$): proton–proton Coulomb repulsion overwhelms the strong nuclear force.
2. **Dirac instability, point nucleus** ($Z \sim 137$): the 1s binding energy reaches mc^2.
3. **Dirac instability, finite nucleus** ($Z \sim 170$): spontaneous e^+e^- pair creation from the vacuum.

The SO($3N$) topology predicts no limit: $\mu_{\text{free}} = 2(N - 2)(N - 1)$ is defined for all N. The physical end of the periodic table is set by nuclear physics ($Z \lesssim 130$), not by electronic topology.

DISCUSSION

Connection to Paper 0's orbital counting

Paper 0 [7] derived the $2n^2$ orbital degeneracy from the geometric packing of nodes on the discrete S^3 lattice. The present result extends this: the S_N representation theory on S^{3N-1} yields not just the orbital counting but the *chemical classification* of atoms. Where Paper 0 derived the $2n^2$ counting from geometric packing, the present analysis provides a group-theoretic framework for understanding why filling those shells produces the periodic pattern C → D → E → B.

Transition metals and Hund's rule

The transition metals (Type E) have partially filled d-shells with ground-state spins $S \geq 1$. The "anomalous" configurations of Cr ($[\mathrm{Ar}]\,3d^5\,4s^1$, $S = 3$) and Cu ($[\mathrm{Ar}]\,3d^{10}\,4s^1$, $S = 1/2$) are often cited as evidence for exchange stabilization. In the present framework, both the "expected" and "anomalous" configurations of Cr have $\nu = 22$ and $\mu_{\mathrm{free}} = 1012$—the topological Level 1 cannot distinguish them. The anomaly lives entirely at Level 2, in the exchange pair count: the half-filled $3d^5\,4s^1$ configuration has 141 exchange pairs versus 136 for $3d^4\,4s^2$, a difference of 5 that tips the energetic balance.

This confirms that Hund's first rule is a *perturbative* effect (exchange stabilization in the Coulomb interaction) rather than a *topological* one (angular momentum reduction on S^{3N-1}). More broadly, the actual chemistry—which electron configuration wins energetically—is determined by exchange, correlation, and orbital-dependent Coulomb effects that operate below the resolution of the topological classification. The Level 1 framework identifies which configurations are *topologically equivalent*; it cannot determine which one is the ground state.

Physical chemistry correspondences

The structure type classification connects naturally to familiar chemical trends:

- **Ionization energy** correlates with the topological cost of surgery on S^{3N-1}. Type B (noble gas) has the highest surgery cost; Type C (alkali) has the lowest.

- **Atomic radius** correlates with the location of the surgery: Type C atoms have a weakly bound outer electron (large radius), while Type B atoms are compact.

- **Electronegativity** increases across a period (C $\to$ B), mirroring the transition from hierarchical to democratic irreps.

- **The periodic law** can be understood as the repetition of the S_N irrep sequence whenever a new principal shell begins to fill.

Fractal structure: core and valence at every scale

A striking feature of the classification is its *scale invariance*. The core–valence decomposition at each ionization step produces the same Types B/C/D structure at every level:

$$
\begin{aligned}
\mathrm{Na} &\to \mathrm{Na}^+ + e^- &\qquad& (\mathrm{C} \to \mathrm{B} + \mathrm{A}),\\
\mathrm{Na}^+ &\to \mathrm{Na}^{2+} + e^- &\qquad& (\mathrm{B} \to \mathrm{C} + \mathrm{A}).
\end{aligned}
$$

Each ionization peels off one column from the Young diagram, converting one type into another in a predictable sequence. This self-similar structure is a direct consequence of the recursive nature of $S_N \supset S_{N-1}$ representations.

CONCLUSION

We have shown that the group structure of the periodic table—the classification of elements into noble gases, alkali metals, alkaline earths, and open-shell atoms—follows from the S_N representation theory of fermionic wavefunctions on the hypersphere S^{3N-1}. The key results are:

1. The ground-state angular quantum number $\nu = N - 2$ is universal for $S < N/2$, yielding the Pauli centrifugal cost $\mu_{\mathrm{free}} = 2(N-2)(N-1)$.

2. The shape of the S_N spatial irrep classifies atoms into types (A/B/C/D/E) that recast the chemical group structure in representation-theoretic language.

3. The periodic law corresponds to the repetition of the irrep sequence C $\to$ D $\to$ E $\to$ B within each period.

4. Hund's first rule and configuration anomalies (Cr, Cu) are Level 2 perturbative effects, invisible to the Level 1 topology.

5. The Dirac instability at $Z\alpha \to 1$ is a metric singularity on S^{3N-1}, not a topological one; the representation theory is smooth through all N.

The approach separates the periodic table into a universal *topological skeleton* (Level 1: ν, μ_{free}, structure type) and a chemistry-specific *metric flesh* (Level 2: Z, α, Coulomb integrals). This separation provides a group-theoretic framework for the periodic table's structure (topology) while leaving the element-specific chemistry to perturbative effects (metric). The framework recasts, rather than derives from first principles, the empirical periodic pattern—its value lies in unifying the classification under a single mathematical structure.

This work uses the hyperspherical formalism developed in the Geometric Vacuum program (Papers 0–15). The S_N analysis was informed by the classical references of Avery [4] and Fock [2, 3].

* Paper 16 in the Geometric Vacuum series

[1] D. I. Mendeleev, *Zh. Russ. Khim. Obshch.* **1**, 60 (1869).
[2] V. Fock, *Z. Phys.* **98**, 145 (1935).
[3] V. Fock, *Izv. Akad. Nauk SSSR, Ser. Fiz.* **18**, 161 (1954).
[4] J. Avery, *Hyperspherical Harmonics and Generalized Sturmians* (Kluwer, Dordrecht, 2000).
[5] H. Weyl, *The Classical Groups: Their Invariants and Representations* (Princeton University Press, Princeton, 1946).
[6] W. Fulton and J. Harris, *Representation Theory: A First Course* (Springer, New York, 1991).
[7] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2025).
[8] J. Loutey, "The Geometric Atom: Atomic Structure as a Packing Problem," GeoVac Paper 1 (2025).
[9] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).
[10] J. Loutey, "The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs," GeoVac Paper 13 (2026).